Forecasting Exchange Rates Using AI tools (Copilot and Gemini).

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Econ 5041 | April 2026

Abstract

This paper examines how well exchange rate forecasts from two AI tools, Microsoft Copilot and Google Gemini, perform in real life. It specifically examines five currencies: the Euro, British Pound Sterling, Norwegian Kroner, Swedish Kroner, and Swiss Franc, with forecasts ranging from 1 to 6 days ahead. Forecasts were gathered over five business days using two different prompts. To assess their performance, I use measures like Average Forecast Error, Root Mean Square Prediction Error, Mean Directional Error, and the Diebold-Mariano-West test to compare them against standard models like random walk and AR(1).

The results show that both AI tools perform reasonably well for stable currencies like EUR and GBP, but struggle more for NOK and SEK. Gemini usually outperforms Copilot under Prompt 1, while Copilot improves significantly under Prompt 2. Overall, neither model beats the simple benchmarks, which aligns with findings from past research.

1. Introduction

Exchange rates move in response to many different economic and market factors, so their behavior is often uneven and hard to predict. These economic factors include interest rates, inflation, policy decisions, and unexpected global events. This has made exchange rate forecasting an important topic in economics, especially because even simple benchmark models have often been difficult to beat in practice (Meese and Rogoff, 1983).

The objective of this paper is to test whether AI-based forecasts from Copilot and Gemini are unbiased and whether they can outperform benchmark models such as the random walk and AR (1). To do this, I collect real-time forecasts for five currencies (EUR, GBP, NOK, SEK, and CHF) over horizons from 1 to 6 days ahead, using two different prompts. The forecasts are evaluated using measures such as Average Forecast Error (AFE), Root Mean Squared Prediction Error (RMSPE), Mean Directional Error, and the Diebold–Mariano–West test, which is used to formally compare whether the forecasting accuracy of the AI models is significantly different from that of the benchmark models.

Furthermore, there is growing research on using artificial intelligence for exchange rate forecasting, but the findings are not always consistent across studies. Qureshi et al. (2023**)** show that artificial intelligence can be used together with time series models to forecast exchange rates and improve prediction accuracy based on different evaluation measures. However, Tsuji (2025) finds that even within AI-based approaches, performance can differ, as some methods produce more accurate forecasts than others depending on the currency being studied. This suggests that while AI has strong potential in exchange rate forecasting, its performance is not always stable and can vary across different situations.

This paper adds to the discussion by comparing two AI tools using real-time data and different prompts. The main findings indicate that both AI tools perform reasonably well for stable currencies like EUR and GBP, where forecast errors are generally small. However, they struggle

more with volatile currencies like NOK and SEK. Additionally, neither tool consistently outperforms the benchmark models. Finally, the results suggest that performance varies based on the currency, the forecast horizon, and also on the prompt used, as forecasts tend to improve when more context is provided.

2. Methods, Models and Data

2.1 Hypothesis

In this paper, I test three null hypotheses. The first null hypothesis is , where is the forecast error at the horizon . In this study, I assess this mainly using the Average Forecast Error (AFE), where values close to zero indicate unbiasedness.

In the second null hypothesis, , I test that the AI forecasts are not more accurate than the benchmark models. To examine this, I use the Diebold–Mariano–West (DMW) test. This test compares the squared forecast errors from the two models.

Finally, my third null hypothesis is that there is no systematic difference in the forecast accuracy between the AI. This is defined as . Here, and represent the forecast errors from Copilot and Gemini, respectively.

2.2 Forecasting and Benchmark Models

The models used in this study include two AI tools, Microsoft Copilot and Google Gemini, and two benchmark models, the Random Walk (RW) and AR (1). The Random Walk assumes that the best forecast is simply the current exchange rate. The AR (1) model builds on this by allowing the exchange rate to depend on its past value.

The AI forecasts are generated using two prompts. Prompt 1 is a simple request for exchange rate forecasts without additional context, while Prompt 2 includes the most recent observed exchange rate as a baseline point. All exchange rates are expressed as domestic currency per USD to ensure consistency across currencies. Where necessary, the data is converted into this form.

2.3 Data

The actual exchange rate data were obtained from the Federal Reserve Bank of St. Louis (FRED) database using daily observations. The sample spanned from April 2021 to April 2026. This longer time period was used mainly to allow the AR(1) model to properly capture persistence in the exchange rate series.

All exchange rates were expressed as domestic currency per USD to ensure consistency across currencies. For some currencies, particularly EUR and GBP, the data was originally reported in the opposite form (USD per currency unit), so it was converted accordingly.

Forecast data collection started on 9th March 2026, and forecasts were collected consistently over five consecutive business days. For each day, new chats were used for each prompt to avoid any carry-over effects from previous interactions. Forecasts that fell on weekends were aligned with the next business day to match the available actual data.

3.Results

3.1 Average Forecast Errors

The AFE results reported in Tables 1–4 show different patterns across currencies and horizons. For EUR, GBP, and CHF, the values are very small (mostly between -0.01 and 0.01) and don't show any clear pattern, suggesting that forecasts for these currencies are roughly unbiased.

For SEK and NOK, the pattern is quite different. Under Prompt 1, Copilot produces large negative AFEs, meaning it consistently underpredicts the exchange rate. For SEK, the AFE ranges from about -1.36 at H=1 to -1.31 at H=6. Gemini also underpredicts, by a margin of about -0.6.

Under Prompt 2, both tools improve, with Copilot's AFE turning positive and much smaller, suggesting that additional context helps improve the forecasts.

3.2 Root Mean Squared Prediction Error (RSMPE)

The RMSPE results in Tables 5–8 show that forecast errors increase with the horizon for all currencies and models, which is expected since uncertainty grows the further ahead, we predict. For EUR and GBP, RMSPE values are small and quite similar across both tools, so there is not much difference in performance. Under Gemini Prompt 1, CHF is slightly different. It records RMSPE values of around 0.07, which are higher than Copilot's.

For SEK and NOK, the differences are more noticeable. Under Prompt 1, Copilot has the largest errors, with RMSPE for SEK starting at about 1.375 at H=1. Gemini performs better, with values below 0.84. Under Prompt 2, Copilot improves and becomes competitive with Gemini.

3.3 Directional Accuracy using Mean Directional Error

The MDE results in Tables 9–10 show how well each tool predicts the direction of exchange rate movements, where lower values indicate better accuracy. Under Prompt 1, Gemini performs very well, with an MDE of 0.000 for SEK, NOK, and CHF, meaning it correctly predicts the direction every time for these currencies. Copilot, in comparison, has higher MDE values (around 0.25 for SEK and NOK), showing more directional errors.

Under Prompt 2, both tools look more similar, with MDE values around 0.5 for most currencies. Overall, Gemini still seems to have a slight edge in directional accuracy, especially under the first prompt.

3.4 Diebold Mariano- West Tests (DMW)

The DMW test results in Tables 11–14 compare the accuracy of the AI forecasts with the AR(1) and random walk benchmarks. For EUR, GBP, and CHF, p-values are all above 0.05 for both tools, so we cannot reject the null that their forecasts are as accurate as the benchmarks. This is not too surprising given how difficult it is to beat simple models for major currencies.

For SEK and NOK, the results are quite different. Under Prompt 1, Copilot performs much worse, with p-values of 0.00 across all horizons, showing its forecasts are significantly less

accurate than the benchmarks. Gemini performs better, with p-values around 0.07– 0.08 for SEK and 0.18–0.24 for NOK, suggesting it comes closer to benchmark performance.

3.5 Unbiasedness Tests

The regression-based unbiasedness tests, where forecast errors are regressed on a constant, were carried out for the AR(1) and random walk benchmark models. The results show that for all currencies, the constant is not statistically significant at the 5% level. This means we cannot reject the null hypothesis of unbiasedness for either benchmark model. The R-squared values are all close to zero, which suggests that there is no clear pattern in the benchmark forecast errors.

For the AI forecasts, the AFE tables suggest a different picture. Under Prompt 1, both Copilot and Gemini would likely fail an unbiasedness test for SEK and NOK, given the large and consistent negative errors. Under Prompt 2, the errors are smaller and more mixed in sign, which makes unbiasedness more plausible for most currencies.

3.6 Panel Regression: Copilot vs Gemini

To compare Copilot and Gemini more formally, I estimate a panel regression where the dependent variable is the difference in squared percent errors (SPE). A positive coefficient means Gemini performs better, while a negative one means Copilot performs better. Newey–West standard errors are used to account for autocorrelation, making the reported significance levels more reliable.

Under Prompt 1, day effects for days 2, 3, and 5 are positive and significant, suggesting Gemini has lower errors on those days. Horizon effects are small and not significant, meaning performance does not vary much across horizons. For currencies, Gemini performs better for SEK and NOK, while Copilot performs better for CHF, with no clear difference for GBP.

Under Prompt 2, Gemini performs better across most days. However, Copilot improves for SEK, while NOK shows weaker effects. Overall, neither model consistently dominates.

3.7 Conclusion

This paper evaluated real-time AI-based exchange rate forecasts from Microsoft Copilot and Google Gemini across five currencies and six forecast horizons. Overall, both tools perform reasonably well for more stable currencies such as EUR, GBP, and CHF, where forecast errors are small and roughly unbiased. However, performance is weaker for more volatile currencies like NOK and SEK, especially under Prompt 1, where Copilot shows underpredictions. Gemini performs better in these cases, but still does not clearly outperform simple benchmark models.

Importantly, comparing the AI forecasts to the random walk and AR(1) benchmarks does not change the overall conclusion. The DMW results show that neither AI tool consistently provides more accurate forecasts than these simple models. This is in line with the strong persistence observed in the data, where the AR(1) model performs well, making it difficult for more complex models to improve on it.

Overall, the results show that AI forecasting performance depends heavily on the prompt used and the type of currency being forecasted. While AI tools can produce reasonable forecasts, they

do not consistently outperform simple benchmarks, and their performance is not stable across settings.

One limitation of this study is the short evaluation period of only five business days, which limits the reliability of statistical inference. A second limitation is that the results depend on the specific prompts used, and small changes in how the task is phrased may lead to different forecasts. Future work could extend the sample period and explore more structured prompting approaches to better assess the potential of AI in exchange rate forecasting.

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Appendix

A. Summary of all tables showing Average Forecast Errors (Table 1-4) and RSMPE (Table 5-8) , MDE (Tables 9 and 10) DMV tests (Tables 11-14)

Table 1: AFE – Copilot, Prompt 1

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.001	0.001	0.001	0.000	-0.002	-0.005
GBP	0.001	0.002	0.002	0.002	0.001	-0.000
SEK	-1.362	-1.328	-1.301	-1.295	-1.307	-1.310
NOK	-0.915	-0.906	-0.913	-0.936	-0.966	-0.954
CHF	0.003	0.004	0.005	0.006	0.005	0.004

Table 2: AFE – Gemini, Prompt 1

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.000	0.001	0.001	0.002	-0.000	-0.003
GBP	-0.001	-0.002	-0.000	-0.001	-0.002	-0.004
SEK	-0.646	-0.629	-0.602	-0.598	-0.616	-0.628
NOK	-0.292	-0.304	-0.317	-0.345	-0.383	-0.380
CHF	-0.054	-0.054	-0.052	-0.050	-0.051	-0.053

Table 3: AFE – Copilot, Prompt 2

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.006	0.009	0.011	0.012	0.012	0.012
GBP	0.004	0.006	0.008	0.008	0.008	0.008
SEK	0.128	0.176	0.217	0.238	0.243	0.257
NOK	0.078	0.095	0.100	0.091	0.094	0.104
CHF	0.007	0.009	0.012	0.015	0.015	0.016

Table 4: AFE – Gemini, Prompt 2

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.003	0.004	0.004	0.003	0.001	-0.002
GBP	0.002	0.002	0.002	0.001	0.001	-0.001

SEK	-0.271	-0.248	-0.233	-0.236	-0.253	-0.267
NOK	-0.117	-0.120	-0.139	-0.171	-0.206	-0.203
CHF	0.004	0.006	0.008	0.008	0.006	0.006

Table 5: RMSPE – Copilot, Prompt 1

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.006	0.008	0.010	0.011	0.012	0.014
GBP	0.005	0.005	0.004	0.004	0.001	0.002
SEK	1.375	1.337	1.307	1.299	1.311	1.317
NOK	0.924	0.916	0.925	0.950	0.990	0.997
CHF	0.005	0.007	0.007	0.007	0.008	0.008

Table 6: RMSPE – Gemini, Prompt 1

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.003	0.005	0.006	0.006	0.005	0.006
GBP	0.008	0.010	0.008	0.008	0.006	0.006
SEK	0.834	0.817	0.827	0.804	0.807	0.836
NOK	0.520	0.524	0.542	0.525	0.569	0.608
CHF	0.072	0.071	0.071	0.069	0.069	0.071

Table 7: RMSPE – Copilot, Prompt 2

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.009	0.012	0.017	0.019	0.022	0.025
GBP	0.007	0.009	0.014	0.015	0.018	0.022
SEK	0.158	0.202	0.258	0.280	0.280	0.325
NOK	0.087	0.137	0.196	0.225	0.215	0.231
CHF	0.009	0.012	0.017	0.020	0.022	0.026

Table 8: RMSPE – Gemini, Prompt 2

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.006	0.007	0.008	0.008	0.010	0.012

GBP	0.005	0.007	0.007	0.006	0.008	0.009
SEK	0.812	0.788	0.754	0.703	0.728	0.772
NOK	0.389	0.389	0.391	0.354	0.374	0.413
CHF	0.007	0.010	0.010	0.011	0.012	0.014

Table 9: Mean Directional Error – Copilot

Currency	Prompt 1	Prompt 2
EUR	0.750	0.750
GBP	0.500	0.500
SEK	0.250	0.750
NOK	0.250	0.750
CHF	0.750	0.750

Table 10: Mean Directional Error – Gemini

Currency	Prompt 1	Prompt 2
EUR	0.250	0.500
GBP	0.750	0.500
SEK	0.000	0.500
NOK	0.000	0.500
CHF	0.000	0.750

Table 11: DMW Test P-Values – Copilot vs. AR(1)

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.40	0.51	0.58	0.50	0.43	0.43
GBP	0.46	0.63	0.27	0.13	0.05	0.21
SEK	0.00	0.00	0.00	0.00	0.00	0.00
NOK	0.00	0.00	0.00	0.00	0.01	0.01
CHF	0.48	0.69	0.76	0.57	0.85	0.89

Table 12: DMW Test P-Values – Copilot vs. Random Walk

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.42	0.57	0.66	0.58	0.48	0.46
GBP	0.47	0.53	0.26	0.10	0.06	0.13
SEK	0.00	0.00	0.00	0.00	0.00	0.00
NOK	0.00	0.00	0.00	0.00	0.01	0.01
CHF	0.49	0.81	0.65	0.40	0.77	0.76

Table 13: DMW Test P-Values – Gemini vs. AR(1)

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.16	0.42	0.36	0.43	0.62	0.87
GBP	0.35	0.45	0.68	0.61	0.39	0.43
SEK	0.08	0.08	0.08	0.08	0.07	0.08
NOK	0.24	0.23	0.21	0.21	0.19	0.18
CHF	0.08	0.08	0.08	0.08	0.07	0.07

Table 14: DMW Test P-Values – Gemini vs. Random Walk

Currency	H=1	H=2	H=3	H=4	H=5	H=6
EUR	0.17	0.39	0.34	0.36	0.49	0.82
GBP	0.35	0.47	0.75	0.72	0.56	0.62
SEK	0.08	0.08	0.08	0.08	0.08	0.08
NOK	0.24	0.23	0.21	0.20	0.19	0.18
CHF	0.08	0.08	0.08	0.08	0.07	0.07

The tables below show the results of the AR(1) model across forecast horizons 1 to 6, illustrating how well past exchange rates predict future values. Overall, the results indicate that exchange rates are highly persistent, as the lagged values have coefficients very close to one and are statistically significant at all horizons

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AR1 Model - Horizon 1
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	Dependent variable:				
	Actual				
	CHF	NOK	GBP	EUR	SEK
	(1)	(2)	(3)	(4)	(5)
Lag 1 Actual	0.998***	0.995***	0.995***	0.995***	0.995***
	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Constant	0.002	0.055*	0.004	0.005	0.049
	(0.002)	(0.027)	(0.002)	(0.003)	(0.026)
Observations	1,196	1,196	1,196	1,196	1,196
R2	0.993	0.991	0.988	0.991	0.992
Adjusted R2	0.993	0.991	0.988	0.991	0.992
Residual Std. Error (df = 1194)	0.004	0.076	0.004	0.004	0.071
F Statistic (df = 1; 1194)	175,324.700***	138,137.200***	101,793.500***	129,522.900***	150,121.900***
Note:					*p<0.05; **p<0.01; ***p<0.001

AR1 Model - Horizon 2

	Dependent variable:				
	Actual				
	CHF	NOK	GBP	EUR	SEK
	(1)	(2)	(3)	(4)	(5)
Lag 1 Actual	0.996***	0.989***	0.990***	0.990***	0.990***
	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Constant	0.004	0.108**	0.008*	0.009*	0.096**
	(0.003)	(0.038)	(0.003)	(0.004)	(0.036)
Observations	1,195	1,195	1,195	1,195	1,195
R2	0.986	0.983	0.977	0.982	0.985
Adjusted R2	0.986	0.983	0.977	0.982	0.985
Residual Std. Error (df = 1193)	0.006	0.107	0.006	0.006	0.099
F Statistic (df = 1; 1193)	83,641.470***	69,467.610***	50,927.420***	64,730.970***	76,023.770***
Note:					*p<0.05; **p<0.01; ***p<0.001

AR1 Model - Horizon 3

	Dependent variable:				
	Actual				
	CHF	NOK	GBP	EUR	SEK
	(1)	(2)	(3)	(4)	(5)
Lag 1 Actual	0.994***	0.985***	0.984***	0.986***	0.986***
	(0.004)	(0.005)	(0.005)	(0.005)	(0.004)
Constant	0.005	0.152***	0.012**	0.013**	0.141**
	(0.004)	(0.046)	(0.004)	(0.004)	(0.043)
Observations	1,194	1,194	1,194	1,194	1,194
R2	0.979	0.975	0.966	0.973	0.977
Adjusted R2	0.979	0.975	0.966	0.973	0.977
Residual Std. Error (df = 1192)	0.008	0.129	0.007	0.007	0.120
F Statistic (df = 1; 1192)	56,371.490***	47,459.920***	34,198.890***	43,396.990***	51,771.510***
Note:					*p<0.05; **p<0.01; ***p<0.001

AR1 Model - Horizon 4

	Dependent variable:				
	Actual				
	CHF	NOK	GBP	EUR	SEK
	(1)	(2)	(3)	(4)	(5)
Lag 1 Actual	0.991***	0.979***	0.977***	0.981***	0.981***
	(0.005)	(0.005)	(0.006)	(0.005)	(0.005)
Constant	0.007	0.216***	0.018***	0.018***	0.195***
	(0.004)	(0.052)	(0.005)	(0.005)	(0.050)
Observations	1,193	1,193	1,193	1,193	1,193
R2	0.973	0.968	0.956	0.965	0.970
Adjusted R2	0.973	0.968	0.956	0.965	0.970
Residual Std. Error (df = 1191)	0.009	0.148	0.008	0.008	0.138
F Statistic (df = 1; 1191)	43,072.500***	35,695.510***	26,109.460***	32,965.740***	38,694.400***

Note: *p<0.05; **p<0.01; ***p<0.001

AR1 Model - Horizon 5

	Dependent variable:				
	Actual				
	CHF	NOK	GBP	EUR	SEK
	(1)	(2)	(3)	(4)	(5)
Lag 1 Actual	0.990***	0.973***	0.975***	0.978***	0.977***
	(0.005)	(0.006)	(0.007)	(0.006)	(0.006)
Constant	0.009	0.272***	0.020***	0.020***	0.230***
	(0.005)	(0.058)	(0.005)	(0.005)	(0.055)
Observations	1,197	1,197	1,197	1,197	1,197
R2	0.967	0.960	0.948	0.957	0.963
Adjusted R2	0.967	0.960	0.948	0.957	0.963
Residual Std. Error (df = 1195)	0.010	0.163	0.009	0.009	0.152
F Statistic (df = 1; 1195)	34,813.190***	29,006.670***	21,778.370***	26,667.290***	31,400.050***

Note: *p<0.05; **p<0.01; ***p<0.001

AR1 Model - Horizon 6

	Dependent variable:				
	Actual				
	CHF	NOK	GBP	EUR	SEK
	(1)	(2)	(3)	(4)	(5)
Lag 1 Actual	0.987***	0.968***	0.969***	0.973***	0.973***
	(0.006)	(0.006)	(0.007)	(0.007)	(0.006)
Constant	0.011*	0.324***	0.025***	0.025***	0.277***
	(0.005)	(0.062)	(0.006)	(0.006)	(0.060)
Observations	1,191	1,191	1,191	1,191	1,191
R2	0.960	0.954	0.939	0.949	0.956
Adjusted R2	0.960	0.954	0.939	0.949	0.956
Residual Std. Error (df = 1189)	0.011	0.176	0.010	0.010	0.166
F Statistic (df = 1; 1189)	28,898.630***	24,472.910***	18,402.080***	22,235.530***	26,053.490***

Note: *p<0.05; **p<0.01; ***p<0.001

The tables below show the Average Forecast Error (AFE) results for the AR(1) and Random Walk models across different currencies and forecast horizons, testing whether the forecast errors are equal to zero (i.e., whether the models are unbiased).

AFE From RW1 Forecast Of CHF

	Dependent variable:					
	AFE Horizon1 (1)	AFE Horizon2 (2)	AFE Horizon3 (3)	AFE Horizon4 (4)	AFE Horizon5 (5)	AFE Horizon6 (6)
Constant	-0.0001	-0.0002	-0.0002	-0.0003	-0.0003	-0.0005
	(0.0001)	(0.0002)	(0.0002)	(0.0003)	(0.0003)	(0.0003)
Observations	1,196	1,195	1,194	1,193	1,197	1,191
R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.004 (df = 1195)	0.01 (df = 1194)	0.01 (df = 1193)	0.01 (df = 1192)	0.01 (df = 1196)	0.01 (df = 1190)
Note:						*p<0.05; **p<0.01; ***p<0.001

AFE From AR1 Forecast Of CHF

	Dependent variable:					
	AFE Horizon1 (1)	AFE Horizon2 (2)	AFE Horizon3 (3)	AFE Horizon4 (4)	AFE Horizon5 (5)	AFE Horizon6 (6)
Constant	-0.01	-0.01	-0.02	-0.03*	-0.03*	-0.04*
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)
Observations	1,197	1,197	1,197	1,197	1,202	1,197
R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.24 (df = 1196)	0.34 (df = 1196)	0.42 (df = 1196)	0.48 (df = 1196)	0.54 (df = 1201)	0.59 (df = 1196)
Note:						*p<0.05; **p<0.01; ***p<0.001

AFE From AR1 Forecast Of GBP

	Dependent variable:					
	AFE Horizon1 (1)	AFE Horizon2 (2)	AFE Horizon3 (3)	AFE Horizon4 (4)	AFE Horizon5 (5)	AFE Horizon6 (6)
Constant	-0.0001	-0.0002	-0.0003	-0.0004	-0.001	-0.001
	(0.0002)	(0.0002)	(0.0003)	(0.0003)	(0.0004)	(0.0004)
Observations	1,197	1,197	1,197	1,197	1,202	1,197
R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.01 (df = 1196)	0.01 (df = 1196)	0.01 (df = 1196)	0.01 (df = 1196)	0.01 (df = 1201)	0.01 (df = 1196)
Note:						*p<0.05; **p<0.01; ***p<0.001

AFE From RW1 Forecast Of GBP

	Dependent variable:					
	AFE Horizon1 (1)	AFE Horizon2 (2)	AFE Horizon3 (3)	AFE Horizon4 (4)	AFE Horizon5 (5)	AFE Horizon6 (6)
Constant	-0.0000	0.0001	0.0001	0.0002	0.0002	0.0002
	(0.0001)	(0.0002)	(0.0002)	(0.0002)	(0.0003)	(0.0003)
Observations	1,196	1,195	1,194	1,193	1,197	1,191

R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.004 (df = 1195)	0.01 (df = 1194)	0.01 (df = 1193)	0.01 (df = 1192)	0.01 (df = 1196)	0.01 (df = 1190)

Note: *p<0.05; **p<0.01; ***p<0.001

AFE From AR1 Forecast Of EUR

	Dependent variable:					
	AFE Horizon1	AFE Horizon2	AFE Horizon3	AFE Horizon4	AFE Horizon5	AFE Horizon6
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.0000	0.0001	0.0001	0.0001	0.0002	0.0002
	(0.0001)	(0.0002)	(0.0002)	(0.0003)	(0.0003)	(0.0003)
Observations	1,197	1,197	1,197	1,197	1,202	1,197
R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.005 (df = 1196)	0.01 (df = 1196)	0.01 (df = 1196)	0.01 (df = 1196)	0.01 (df = 1201)	0.01 (df = 1196)

Note: *p<0.05; **p<0.01; ***p<0.001

AFE From RW1 Forecast Of EUR

	Dependent variable:					
	AFE Horizon1	AFE Horizon2	AFE Horizon3	AFE Horizon4	AFE Horizon5	AFE Horizon6
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	-0.0000	-0.0000	0.0000	0.0001	0.0002	0.0002
	(0.0001)	(0.0002)	(0.0002)	(0.0002)	(0.0003)	(0.0003)
Observations	1,196	1,195	1,194	1,193	1,197	1,191
R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.004 (df = 1195)	0.01 (df = 1194)	0.01 (df = 1193)	0.01 (df = 1192)	0.01 (df = 1196)	0.01 (df = 1190)

Note: *p<0.05; **p<0.01; ***p<0.001

AFE From AR1 Forecast Of NOK

	Dependent variable:					
	AFE Horizon1	AFE Horizon2	AFE Horizon3	AFE Horizon4	AFE Horizon5	AFE Horizon6
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.0000	-0.0000	0.0000	0.0000	0.0000	0.0000
	(0.002)	(0.003)	(0.004)	(0.004)	(0.005)	(0.01)
Observations	1,196	1,195	1,194	1,193	1,197	1,191
R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.08 (df = 1195)	0.11 (df = 1194)	0.13 (df = 1193)	0.15 (df = 1192)	0.16 (df = 1196)	0.18 (df = 1190)

Note: *p<0.05; **p<0.01; ***p<0.001

AFE From RW1 Forecast Of NOK

	Dependent variable:					
	AFE Horizon1	AFE Horizon2	AFE Horizon3	AFE Horizon4	AFE Horizon5	AFE Horizon6
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.0004	0.001	0.003	0.004	0.01	0.01
	(0.002)	(0.003)	(0.004)	(0.004)	(0.005)	(0.01)
Observations	1,196	1,195	1,194	1,193	1,197	1,191
R2	0.00	0.00	0.00	0.00	0.00	0.00

Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.08 (df = 1195)	0.11 (df = 1194)	0.13 (df = 1193)	0.15 (df = 1192)	0.16 (df = 1196)	0.18 (df = 1190)
Note:						*p<0.05; **p<0.01; ***p<0.001

AFE From AR1 Forecast Of SEK

	Dependent variable:					
	AFE Horizon1 (1)	AFE Horizon2 (2)	AFE Horizon3 (3)	AFE Horizon4 (4)	AFE Horizon5 (5)	AFE Horizon6 (6)
Constant	-0.001	-0.002	-0.003	-0.004	-0.005	-0.01
	(0.002)	(0.003)	(0.004)	(0.004)	(0.005)	(0.01)
Observations	1,197	1,197	1,197	1,197	1,202	1,197
R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.08 (df = 1196)	0.11 (df = 1196)	0.13 (df = 1196)	0.15 (df = 1196)	0.17 (df = 1201)	0.18 (df = 1196)
Note:						*p<0.05; **p<0.01; ***p<0.00

AFE From RW1 Forecast Of SEK

	Dependent variable:					
	AFE Horizon1 (1)	AFE Horizon2 (2)	AFE Horizon3 (3)	AFE Horizon4 (4)	AFE Horizon5 (5)	AFE Horizon6 (6)
Constant	0.0003	0.001	0.002	0.003	0.004	0.005
	(0.002)	(0.003)	(0.003)	(0.004)	(0.004)	(0.005)
Observations	1,196	1,195	1,194	1,193	1,197	1,191
R2	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R2	0.00	0.00	0.00	0.00	0.00	0.00
Residual Std. Error	0.07 (df = 1195)	0.10 (df = 1194)	0.12 (df = 1193)	0.14 (df = 1192)	0.15 (df = 1196)	0.17 (df = 1190)
Note:						*p<0.05; **p<0.01; ***p<0.001

Table 16: Panel Regression Results Comparing Copilot and Gemini (Change in SPE)

Dependent variable:	
Change in SPE	
Prompt 1 (1)	Prompt 2 (2)
2	0.006***	0.010***
	(0.0003)	(0.001)
3	0.001**	0.008***
	(0.0003)	(0.0005)
4	0.0002	0.008***
	(0.0003)	(0.0004)
5	0.005***	0.008***
	(0.0004)	(0.0005)

2	-0.0002	0.0002
	(0.0004)	(0.001)
3	-0.0004	0.001
	(0.0005)	(0.001)
4	-0.0001	0.001
	(0.0005)	(0.001)
5	0.0001	0.001
	(0.001)	(0.001)
6	-0.0002	0.001
	(0.001)	(0.001)
GBP	-0.0002	-0.00004
	(0.0001)	(0.00003)
SEK	0.012***	-0.006
	(0.002)	(0.003)
NOK	0.006***	-0.002*
	(0.001)	(0.001)
CHF	-0.008***	0.00003
	(0.002)	(0.0003)
Constant	-0.002***	-0.007***
	(0.001)	(0.001)
Note:	*p<0.05; **p<0.01; ***p<0.001	

Notes: *Dependent variable is the change in SPE (Copilot − Gemini). Positive values indicate better performance by Gemini. EUR is the reference category. Newey–West standard errors are reported in parentheses. * p<0.05, ** p<0.01, *** p<0.001*